STAT 212 Midterm practice solutions.

1. $X_1, X_2, \ldots \overset{iid}{\sim} \text{Ber}(p)$

By SLLN, $\frac{1}{n}\sum_{i=1}^{n} X_i \xrightarrow{a.s.} p$.

Consider the subset

$$\mathcal{C}_p = \left\{ x \in \mathbb{R}^{\mathbb{N}} : \frac{1}{n}\sum_{i=1}^{n} x_i \longrightarrow p \right\}$$

SLLN $\Rightarrow$ $\mathbb{P}_p(\mathcal{C}_p) = 1$.

The proof is done upon observing that $\mathcal{C}_p \cap \mathcal{C}_q = \varphi$ for $p \neq q$.

2. $\{B^1_t : t \geq 0\}$, $\{B^2_t : t \geq 0\}$ ind sBM.

$X_t = O B_t$.

$$O = \begin{bmatrix} O_{11} & O_{12} \\ O_{21} & O_{22} \end{bmatrix}$$

$$X_t = \begin{bmatrix} X^1_t \\ X^2_t \end{bmatrix}.$$

$$OO^T = O^T O = I.$$

Note:

(i) For any $k \geq 1$ & $t_1 \leq \cdots \leq t_k$,

$(X^1_{t_1}, \ldots, X^1_{t_k})$ and $(X^2_{t_1}, \ldots, X^2_{t_k})$

are linear combinations of multivariate gaussian.

Hence multivariate gaussian.

$\mathbb{E}[X_t] = 0.$

$$\begin{aligned} \operatorname{cov}(X_t, X_s) &= \operatorname{cov}(OB_t, OB_s) \\ &= O \operatorname{cov}(B_t, B_s) O^T \\ &= O \begin{bmatrix} \min\{s,t\} & 0 \\ 0 & \min\{s,t\} \end{bmatrix} O^T \\ &= \min\{s,t\} I. \end{aligned}$$

Hence (X^1_t, X^2_t) are independent gaussian processes with the appropriate mean & covariances.

Finally, $t \mapsto X_t$ is also cont a.s.

(composition of linear map with

cont map)

This establishes that (X_t^1, X_t^2) are ind standard BM.

3. T_1, T_2 are stopping times wrt the right continuous filtration $\{\mathcal{F}_t^+ : t \geq 0\}$.

First, assume T_1 & T_2 assume discrete values $0 \leq t_1 \leq t_2 \leq \cdots$

[We can choose common values by taking the union of the two supports]

$T = T_1 + T_2$ assumes values $\{u_1, u_2, \ldots\}$

For any $t > 0$

$$\{T < t\} = \bigcup_{u_i < t} \{T = u_i\}$$

$$= \bigcup_{u_i < t} \bigcup_{\substack{t_k + t_\ell \\ = u_i}} \{T_1 = t_k, T_2 = t_\ell\} \in \mathcal{F}_t^+.$$

This completes the proof for the discrete case.

Now, approximate

$$T_{1,n} = \frac{m+1}{2^n} \quad \text{if} \quad \frac{m}{2^n} \le T_1 < \frac{m+1}{2^n}$$

$$T_{2,n} = \frac{m+1}{2^n} \quad \text{if} \quad \frac{m}{2^n} \le T_2 < \frac{m+1}{2^n}$$

$T_{1,n}, T_{2,n}$ are stopping times wrt $\{\mathcal{F}_t^+ : t \ge 0\}$

By the discrete result,

$T_{1,n} + T_{2,n}$ is a stopping time wrt $\{\mathcal{F}_t^+ : t \ge 0\}$

Also, $T_{1,n} \downarrow T_1$, $T_{2,n} \downarrow T_2$ as $n \to \infty$.

$$\{T_1 + T_2 \le t\} = \bigcap_{s > t} \{T_1 + T_2 < s\}$$

$$= \bigcap_{s>t} \underbrace{\bigcup_{n\geq 1} \{T_{1,n} + T_{2,n} < s\}}_{\in \mathcal{F}_s^+}$$

$\in \mathcal{F}_t^+$ by the right continuity of the filtration.

This completes the proof.

4. $\{(Y_n, \mathcal{F}_n) : n \geq 0\}$ – UI martingale.

$Y_n \xrightarrow{a.s.} Y_\infty \qquad \mathbb{P}(T < \infty) = 1$

T is a stopping time wrt $\{\mathcal{F}_n : n \geq 0\}$

$\mathcal{F}_T = \{A : A \cap \{T = n\} \in \mathcal{F}_n\}$ – stopped σ-algebra.

① Y_T is $\mathcal{F}_T$-measurable.

<u>Pf</u>: $Y_T = \sum_{n=0}^{\infty} Y_n \mathbb{1}(T = n)$.

For any Borel set $S \subseteq \mathbb{R}$

$$\{Y_T \in S\} = \bigcup_{n \geq 0} \{Y_T \in S, T = n\}$$

$$= \bigcup_{n \geq 0} \{Y_n \in S, T = n\}$$

$$\therefore \{Y_T \in S, T = m\} = \{Y_m \in S, T = m\} \in \mathcal{F}_m.$$

$\Rightarrow \{Y_T \in S\} \in \mathcal{F}_T$ for all borel sets S.

② Fix $A \in \mathcal{F}_T$. To show $\mathbb{E}[Y_T \mathbb{1}_A] = \mathbb{E}[Y_\infty \mathbb{1}_A]$

$$\mathbb{E}[Y_T \mathbb{1}_A] = \sum_{n=0}^{\infty} \mathbb{E}[Y_n \mathbb{1}_{A \cap \{T=n\}}]$$

Note $Y_n = \mathbb{E}[Y_\infty \mid \mathcal{F}_n]$ & $A \cap \{T = n\} \in \mathcal{F}_n$.

$$\mathbb{E}[Y_T \mathbb{1}_A] = \sum_{n=0}^{\infty} \mathbb{E}[Y_\infty \mathbb{1}_{A \cap \{T=n\}}]$$

$$= \mathbb{E}[Y_\infty \mathbb{1}_A]. \quad \text{(Proved)}$$

5. $\{(X_n, \mathcal{F}_n) : n \geq 1\}$ martingale with $\sup \mathbb{E} X_n^2 < \infty$.

$$Y_0 = 0. \qquad Y_n = \sum_{k=1}^{n} a_k (X_k - X_{k-1})$$

$$\sum_k a_k^2 < \infty$$

$$\mathbb{E}[Y_{n+1} | \mathcal{F}_n] = Y_n + a_{n+1} \mathbb{E}[X_{n+1} - X_n | \mathcal{F}_n]$$
$$= Y_n.$$

Thus $\{(Y_n, \mathcal{F}_n) : n \geq 1\}$ is a marting-ale

$$\mathbb{E} Y_n^2 = \mathbb{E}\left[\left(\sum_{k=1}^{n} a_k (X_k - X_{k-1})\right)^2\right.$$
$$= \mathbb{E}\left[\sum_{k=1}^{n} a_k^2 (X_k - X_{k-1})^2\right]$$
$$+ \mathbb{E}\left[\sum_{k \neq k'} a_k a_{k'} (X_k - X_{k-1})(X_{k'} - X_{k'-1})\right.$$

The diagonal terms

$$\sum_{k=1}^{n} a_k^2 \, \mathbb{E}\left[(X_k - X_{k-1})^2\right] \leq 4 \sup_n \mathbb{E} X_n^2 \times \sum_{k=1}^{\infty} a_k^2 .$$

For any off-diagonal term with $k' < k$

$$\mathbb{E}\left[(X_{k'} - X_{k'-1})(X_k - X_{k-1})\right]$$

$$= \mathbb{E}\left[\mathbb{E}\left[(X_{k'} - X_{k'-1})(X_k - X_{k-1}) \,\middle|\, \mathcal{F}_{k-1}\right]\right]$$

$$= 0.$$

Thus $\{(Y_n, \mathcal{F}_n) : n \geq 1\}$ is an L^2-bounded martingale.

Thus it converges a.s. & in L^2.

(Proved)

<u>Note</u>: This is the discrete analogue of a stochastic integral.